BINDURA UNIVERSITY OF SCIENCE EDUCATION

Faculty of Science — Department of Computer Science

Group Project Technical Report

Smart Exam Invigilation: CNN-based Cheating Behaviour Detection

Course: **CS 414 / NWE 414 / SWE 414 — Artificial Intelligence and Intelligent Systems**

Semester: August–December 2026 | Lecturer: Mr. Zano

Document date: 2026-11-07

Project branding: **Group Project** (replaces prior single-reg working title)

Deliverable	Group Project — code, models, technical report (this PDF)
Primary notebook	notebooks/Group_Project_Colab_Fixed.ipynb
Model artefacts	models/best.pt , models/the_best.pt
Shared data (Drive)	https://drive.google.com/drive/folders/1tcj53raozUaS2jx3-BpbKT5ITUB1G1F5
Drive owners (multi)	sehmaluva; kutadzaushenigel24@gmail.com; venontnyadombo@gmail.com
Kaggle video dataset	https://www.kaggle.com/datasets/fairouzfarouq/student-cheating-behavior-dataset
Kaggle image dataset	https://www.kaggle.com/datasets/ardutraagiginting/exam-cheating-dataset
Version control	GitHub repository under account venontn

1. Problem Definition & Design

Manual exam invigilation does not scale to large halls and online/hybrid assessments. This Group Project designs an intelligent vision system that analyses exam-room imagery (and optionally video) to estimate the probability that a student is exhibiting cheating-related behaviour, then flags intervals for human review. The system is positioned as decision support, not autonomous accusation.

Objectives

- Formulate cheating detection as a supervised binary classification problem over person crops.
- Apply transfer learning (EfficientNet) with modern training practices (EMA, MixUp, TTA).
- Evaluate with precision, recall, macro-F1, average precision, and confusion matrices (CLO 4/5).
- Document ethics, bias risks, and human-in-the-loop deployment (CLO 8/9).
- Ship reproducible code, checkpoints, and GitHub documentation as required by the course outline.

Alignment to Course Learning Outcomes

- **CLO 4 / CLO 5:** supervised learning + CNN transfer learning in PyTorch; metrics beyond accuracy.
- **CLO 8:** privacy, false-positive harm, fairness across camera angles/lighting, governance.
- **CLO 9:** team design, implementation, documentation, version control, demonstration.

PEAS (Intelligent Agent View)

Component	Description
Performance	High precision at a target threshold; low false alarms; calibrated scores
Environment	Exam halls / recorded video; partial observability; dynamic lighting
Actuators	Flag events, annotated frames/video, thresholded alerts for invigilators
Sensors	RGB camera frames; YOLO person detections as attention mechanism

2. Literature & Technique Justification

Cheating behaviour recognition is typically posed as action/behaviour classification or detection on classroom imagery. We adopt a two-stage pipeline common in applied computer vision: (1) person localisation with a lightweight YOLO detector; (2) EfficientNet image classifier on crops. EfficientNet provides a strong accuracy/compute trade-off for Colab/TPU/GPU training. Compared with training a detector end-to-end for every behaviour class, this design reuses mature person detection and focuses labelled effort on behaviour labels.

- Transfer learning from ImageNet (CLO 5).
- Class-imbalance handling via weighted cross-entropy and optional MixUp.
- Threshold chosen for target precision to reduce wrongful flags.
- Cross-dataset checks to measure generalisation between image collections.

3. Data Collection, Sources & Preprocessing

Datasets used

- **Shared Drive package (SmartInvigilation):** https://drive.google.com/drive/folders/1tcj53raozUaS2jx3-BpbKT5ITUB1G1F5 — includes ExamCheatingDataset image splits and examdatasetnew behavioural folders (e.g. normal, frontwatching, sidewatching, suspiciouswatching). **Multiple owners:** sehmaluva; kutadzaushenigel24@gmail.com; venontnyadombo@gmail.com.
- **Hashemite student cheating behaviour (Kaggle videos):** https://www.kaggle.com/datasets/fairouzfarouq/student-cheating-behavior-dataset — optional video source; frames sampled for training/evaluation.

- **Exam cheating images (Kaggle):** https://www.kaggle.com/datasets/ardutraagiginting/exam-cheating-dataset — classes such as looking friend, giving code, giving object, normal act, cheating.

Preprocessing pipeline

- Folder-name mapping to binary labels (normal vs cheating), with soft label for ambiguous frontwatching.
- Drop obvious flip/copy duplicate filenames; perceptual-hash near-duplicate grouping.
- YOLOv8n person crops with padding; cache resized crops for training speed.
- Augmentation: random resized crop, flip, rotation, colour jitter, blur, JPEG, downscale-upscale.
- Grouped stratified folds so near-duplicates do not leak across train/validation.

4. Technical Implementation

Implementation is Python/PyTorch on Google Colab (GPU/TPU) with a local verification script. Training uses a frozen-backbone head warm-up then full fine-tune with AdamW, cosine decay, gradient clipping, EMA weights, MixUp, and optional TTA at validation. The exported artefact is **best.pt** containing architecture name, state dict, class names, threshold, and temperature.

Repository layout (Group Project)

```
GroupProject/
  notebooks/Group_Project_Colab_Fixed.ipynb
  notebooks/Group_Project_Training.ipynb
  models/best.pt , models/the_best.pt
  results/verification_report.json
  docs/Group_Project_Technical_Report.pdf
  scripts/verify_results.py
  README.md
```

5. Evaluation & Verified Results

Per the course outline, we report precision, recall, macro-F1, AP and confusion-oriented analysis rather than accuracy alone. Checkpoints provided in Downloads were loaded and executed locally to verify integrity.

File	Arch	Input	Threshold	Temp	Params	Status
best.pt	efficientnet_b0	64	0.379	0.50	4,052,175	PASS
the_best.pt	efficientnet_b0	64	0.379	0.50	4,052,175	PASS

- **best.pt**: export=fold_ema_ensemble_n=2; sample scores=[0.5166, 0.4965, 0.4945, 0.4771, 0.5046, 0.5004, 0.4979, 0.5152]; Smoke/Colab checkpoint verified loadable. input_size=64 indicates a smoke or reduced run; full dataset training uses 224/288.
- **the_best.pt**: export=fold_ema_ensemble_n=2; sample scores=[0.5166, 0.4965, 0.4945, 0.4771, 0.5046, 0.5004, 0.4979, 0.5152]; Smoke/Colab checkpoint verified loadable. input_size=64 indicates a smoke or reduced run; full dataset training uses 224/288.

Interpretation: Both provided weights load successfully and produce finite cheating scores. An input size of 64 indicates these artefacts come from a reduced/smoke training configuration. For the final graded demonstration, re-run `Group_Project_Colab_Fixed.ipynb` on GPU/TPU against the full Drive/Kaggle data to regenerate full-resolution **best.pt** and report OOF precision/recall/F1/AP from that run.

6. Ethical Consideration & Bias Analysis

- False positives can unjustly accuse students — operate at high precision and always keep a human reviewer.
- Camera angle, skin tone under lighting, seating position, and cultural gesture norms can bias scores; soft labels and threshold calibration partially mitigate but do not eliminate risk.
- Store flagged clips with access control; follow institutional exam policy and Zimbabwe data-protection expectations discussed in Week 14 of the outline.

- Declare dataset licences/citations and any AI assistance used in development.
- Do not deploy as sole evidence for disciplinary action.

7. Documentation, GitHub & Drive Ownership

Course requirement 8.1 explicitly mandates documentation and GitHub version control. This Group Project package is prepared for publication under GitHub user **venontn**.

Google Drive note: The shared folder https://drive.google.com/drive/folders/1tcj53raozUaS2jx3-BpbKT5ITUB1G1F5 lists **multiple owners**: **sehmaluva**, **kutadzaushenigel24@gmail.com**, and **venontnyadombo@gmail.com**. It is a classroom resource, not a single-owner submission folder. For graded deliverables, create a personal Drive folder owned by **venontnyadombo@gmail.com**, copy notebook / best.pt / this report into it, and set sharing to Restricted (lecturer view-only if required). Cite the multi-owner dataset folder as a data source only.

8. Demonstration Plan (15 minutes)

- Problem & PEAS (2 min).
- Data sources + preprocessing diagram (3 min).
- Live Colab/local inference on an image with YOLO boxes and FLAG overlay (5 min).
- Metrics discussion and ethics (3 min).
- Q&A; (2 min).

9. Conclusion

The Group Project delivers a reproducible CNN-based exam-behaviour classifier with person-centric crops, calibrated thresholding, and explicit ethical constraints. Verified checkpoints confirm the export format expected by an invigilation backend. Completing a full-data Colab train produces the final submission-quality best.pt alongside this report and GitHub repository.

References

- Russell, S. & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.).
- Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.).
- Tan, M. & Le, Q. (2019). EfficientNet: Rethinking Model Scaling for CNNs.
- BUSE CS 414 / SWE 414 Course Outline — Artificial Intelligence and Intelligent Systems (2026).
- Kaggle: fairouzfarouq/student-cheating-behavior-dataset; ardutraagiginting/exam-cheating-dataset.
- SmartInvigilation shared Drive folder (classroom dataset package).

— End of Group Project Technical Report —